1. Slide Verification Problem #1

Slope, homogenous

1.1. Introduction

In 1988 a set of 5 basic slope stability problems, together with 5 variants, was distributed both in the Australian Geomechanics profession and overseas as part of a survey sponsored by ACADS (Giam & Donald (1989)). This is the ACADS 1(a) problem.

1.2. Problem Description

This problem as shown in Figure 1.1 is the simple case of a total stress analysis without considering pore water pressures. It represents a homogenous slope with soil properties given in Table 1.1. The factor of safety and its corresponding critical circular failure is calculated.

A slip center search grid of 20 x 20 intervals is used, with 11 circles per grid point, generating a total of 4851 circular slip surfaces. Grid is located at (22.8, 62.6), (22.8, 42.3), (43.7, 62.6), and (43.7, 42.3). Tolerance is 0.0001.

1.3. Geometry and Material Properties

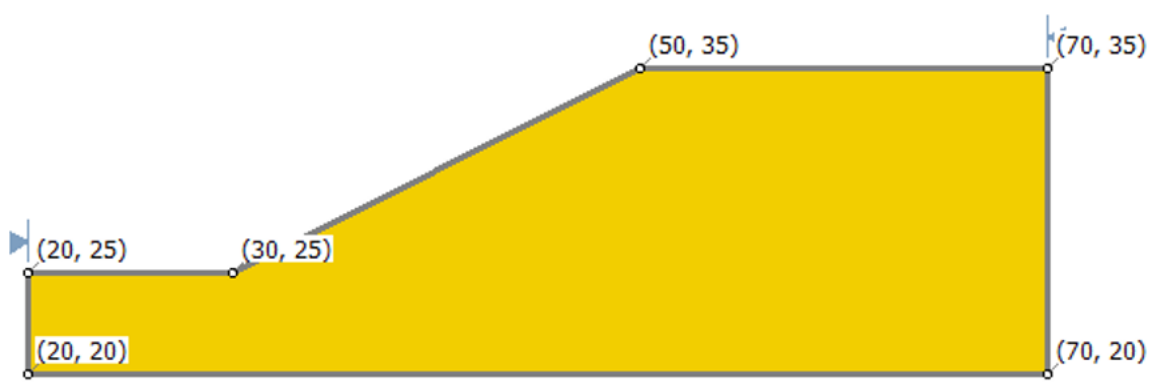

Figure 1.1: Geometry Setup in **Slide**

Table 1.1: Soil Properties

c´ (kN/m^2)	ϕ´ (deg.)	γ (kN/m^3)
3.0	19.6	20.0

1.4. Results

Table 1.2

Method	Factor of Safety
Bishop	0.987
Spencer	0.986
GLE	0.986
Janbu Corrected	0.990

Note: Reference factor of safety = 1.00 [Giam]

Mean Bishop FOS (18 samples) = 0.993

Mean FOS (33 samples) = 0.991

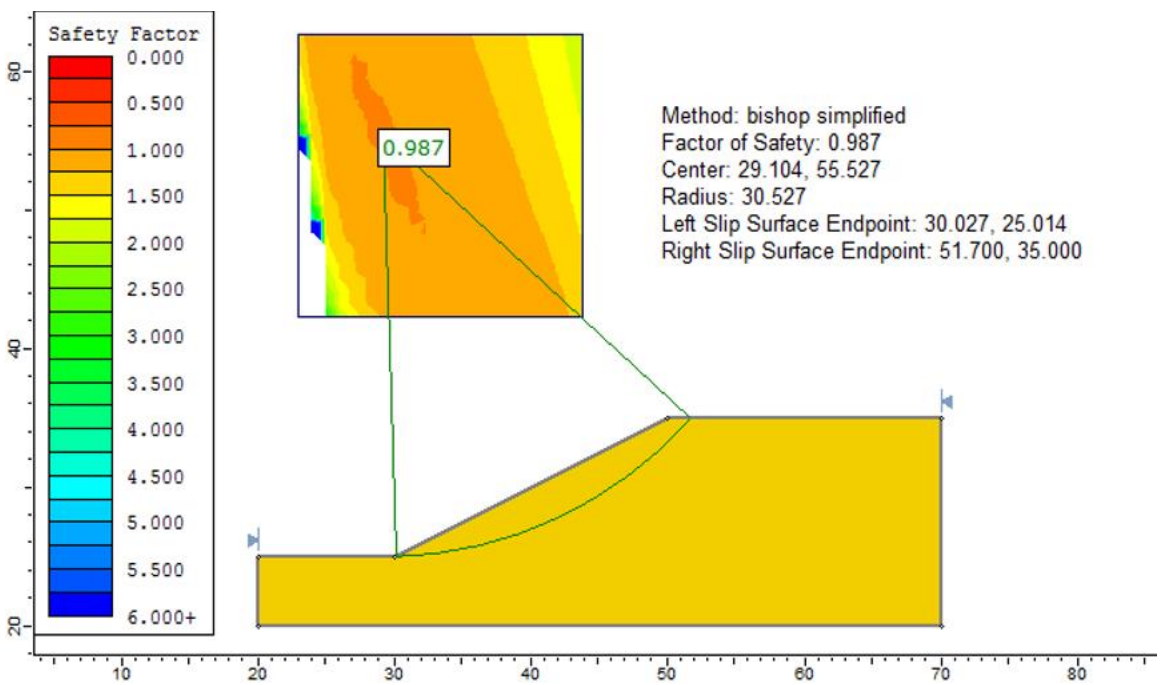

Figure 1.2: Solution, using the Bishop simplified method

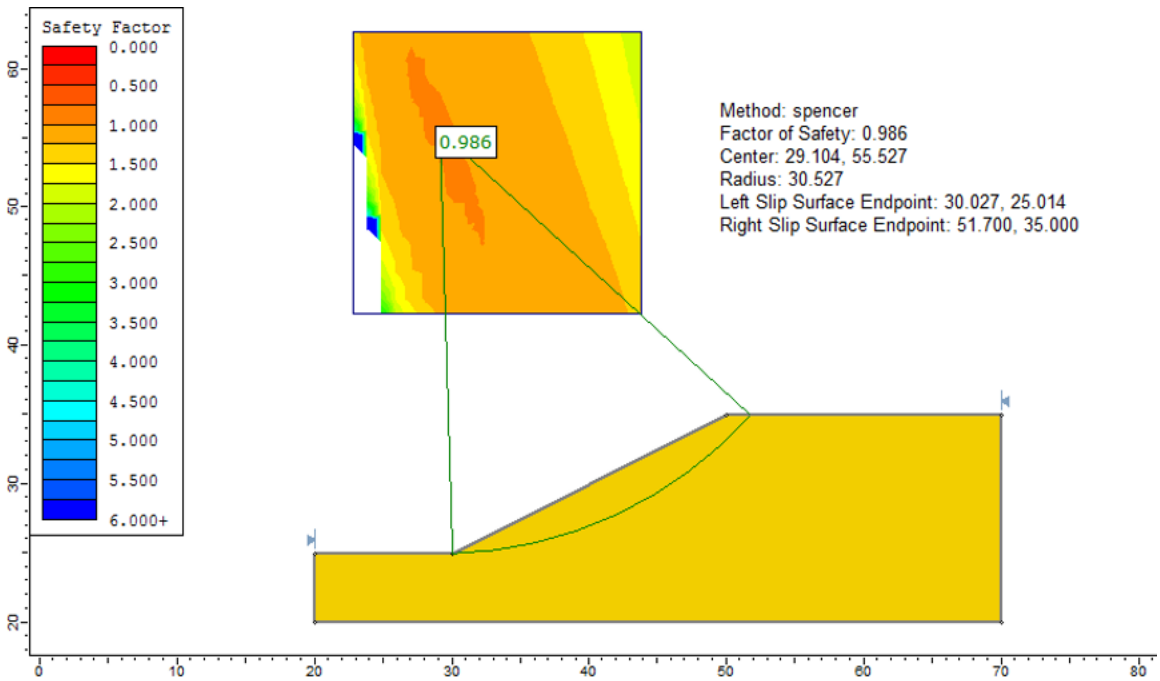

Figure 1.3: Solution, using the Spencer method

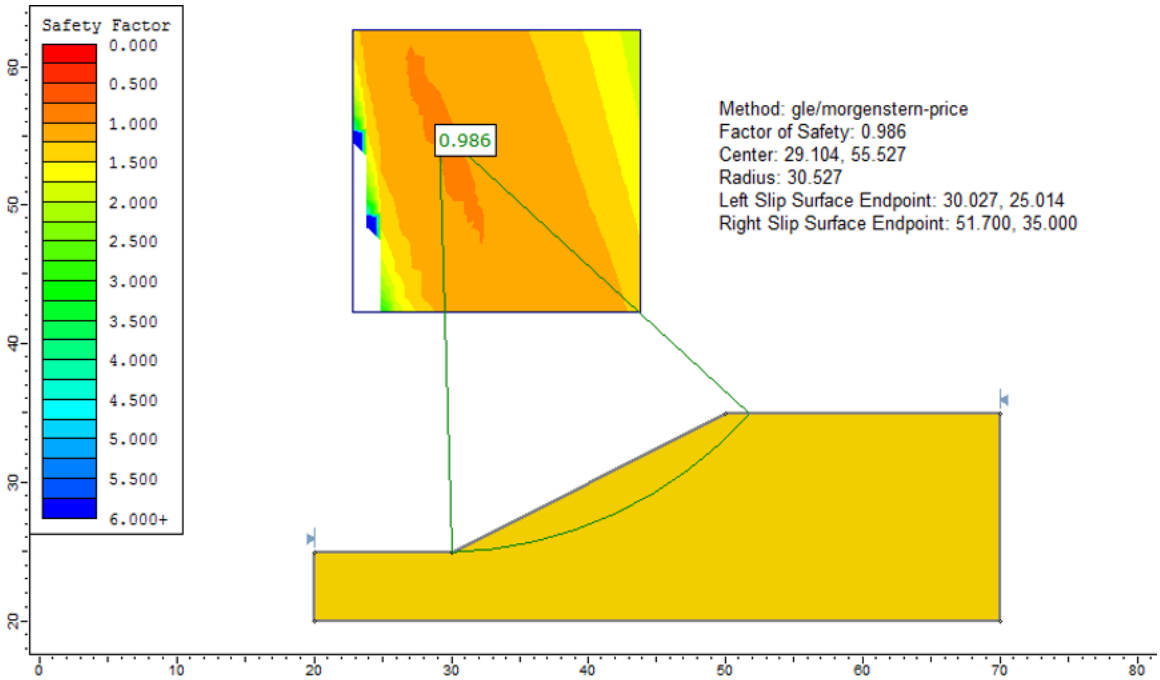

Figure 1.4: Solution, using the GLE method

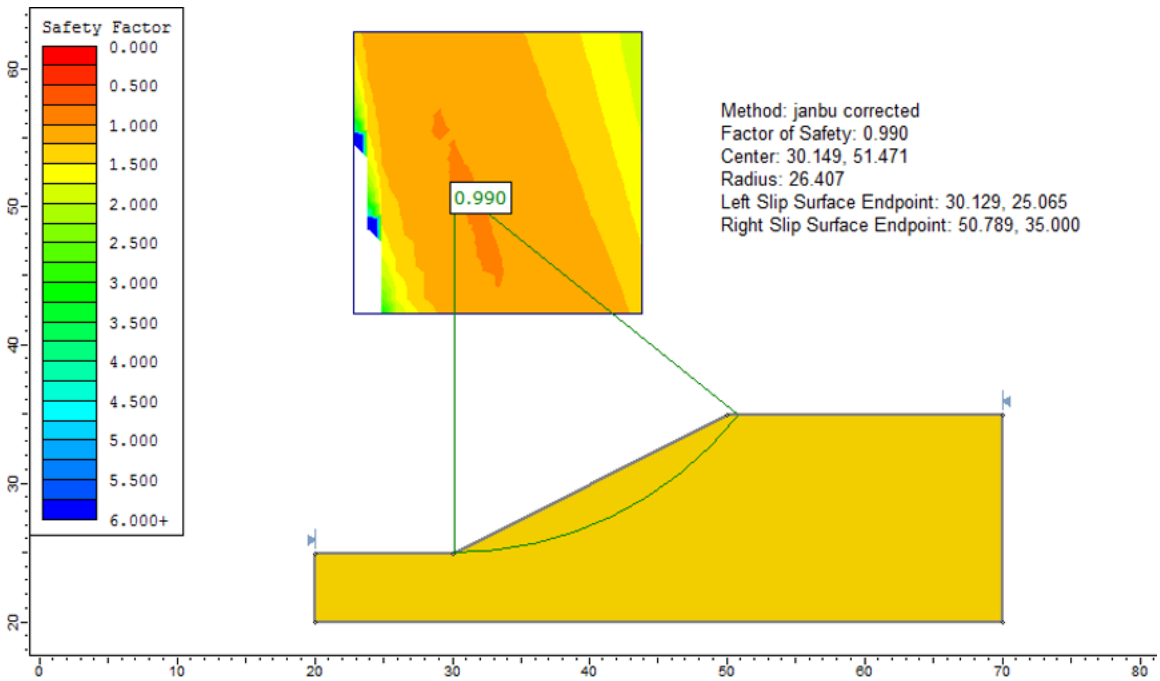

Figure 1.5: Solution, using the Janbu corrected method